Phased-array Compounding for Reflection Ultrasound Computed Tomography Systems

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***Abstract*— This work introduces a phased-array compounding technique to enhance the image quality of Reflection Ultrasound Computed Tomography (RUCT) systems. Conventional RUCT systems often suffer from low Signal-to-Noise Ratio (SNR) and Contrast-to-Noise Ratio (CNR), particularly at off-center locations, due to the use of single-element transmission. Our method utilizes multiple transmit elements with beam steering to distribute acoustic energy more effectively across the entire field of view (FOV). Numerical studies using simulated phantoms demonstrate that the proposed technique improves SNR and CNR by approximately 11 dB and 18 dB, respectively, resulting in high-contrast images across the FOV. This approach offers a significant advancement in RUCT imaging, with potential applications in improving diagnostic accuracy.**

***Keywords*— Reflection Ultrasound Computed Tomography, Phased-array compounding, Beam steering, Image reconstruction.**

I. Introduction

B-mode ultrasound is widely used in medical imaging due to its real-time capabilities, non-invasive nature, and cost-effectiveness. However, it has significant limitations, including low spatial resolution and susceptibility to speckle noise, particularly in deeper tissues [1]. Reflection Ultrasound Computed Tomography (RUCT) addresses these challenges by reconstructing images from multiple reflection angles, offering enhanced spatial resolution and reduced artifacts [2-4]. These advantages make RUCT a promising alternative for more accurate imaging, especially in complex tissue environments where conventional B-mode ultrasound may be inadequate [5].

A typical RUCT system often relies on the synthetic aperture technique to reconstruct images. In this approach, pulses are transmitted sequentially from a single transducer element, and the reflected signals are collected over multiple angles to form a comprehensive image [3, 6]. While the synthetic aperture method enables detailed imaging from various perspectives, it suffers from certain limitations. The most significant limitation is the low signal-to-noise ratio (SNR), as single-element pulse transmission is has limited acoustic power to sonicate the medium which results in weaker signals and increased susceptibility to noise, thus degraded image quality [7]. Additionally, RUCT systems often experience long acquisition times due to the sequential nature of data collection, and the reconstruction process requires substantial computational resources, which can be a bottleneck for real-time imaging [8, 9]. These limitations underscore the need for advanced techniques to improve the performance and clinical viability of RUCT systems.

To overcome the limitations of traditional systems, various approaches have been proposed in the literature. One common strategy involves the use of multi-element transducer arrays, which allow simultaneous transmission and reception of ultrasound waves across multiple elements [10, 11]. This approach can improve the SNR by increasing the effective aperture and reducing noise, leading to clearer images. Additionally, model-based [12], and adaptive beamforming [13] techniques have been explored to enhance image resolution and contrast of the final image. Other studies have investigated the integration of advanced signal processing algorithms, such as coherent compounding [14, 15], weighting factors [6, 16, 17], signal restoration methods [18], iterative-based, and machine learning-based reconstructions [19, 20], to reduce noise and artifacts while speeding up image acquisition and processing. These advancements have shown potential in addressing the inherent challenges of RUCT. However, they also introduce new challenges, such as increased hardware complexity and computational demands, which must be carefully balanced to achieve optimal performance.

In this study, we propose an innovative approach to enhance the SNR in RUCT systems through a modified strategy for acoustic transmission. In this approach, in contrast to conventional RUCT which utilizes emitting pulses from a single transmit element, multiple adjacent elements are used to transmit with zero transmit delays, creating a narrow main-lobe that significantly improves the SNR at the center of the image. However, this approach naturally leads to lower SNR values at off-center regions. To address this, we further enhance image quality across the entire field of view (FOV) by steering the transmit beam in multiple directions with pre-defined transmit delays during consecutive transmits. This technique ensures a uniformly high-quality RUCT image, offering a robust solution to the limitations of traditional RUCT systems.

II. Materials and Methods

A. System Setup

Numerical studies were conducted using the k-Wave® toolbox [21]. The simulation environment was configured to mimic the setup of a RUCT system.

In our simulation studies, 128 transmit elements (TEs) were arranged in a circular array with a radius of 40 mm, surrounding the region of interest. Each TE operated at a transmit frequency of 2 MHz with a pulse duration of 2 cycles. The circular arrangement of the TEs allowed for comprehensive coverage of the imaging region, ensuring that the ultrasound waves could be transmitted and received from

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multiple angles. For each transmit event, 30 TEs were utilized to capture and reconstruct the RUCT images. The specific configuration of TEs and their operational parameters were chosen to optimize the system's imaging capabilities while maintaining computational efficiency.

B. Single- and Multi-Element Transmission Technique

1) Single-Element Transmission (Conventional Method): In RUCT systems, ultrasound pulses are typically transmitted using a single element at each transmitting stage. While this method is quite straightforward, it has significant limitations, particularly in terms of the acoustic energy imparted to the region of interest (ROI). As illustrated in Fig. 1(a), when a single element transmits an ultrasound pulse, the induced acoustic energy is relatively low, leading to a weak signal that is susceptible to noise. This limitation can result in low SNR and reduced image quality, especially in deeper or more complex tissues.

2) Multi-Element Transmission Technique: To enhance the acoustic energy and improve SNR, a multi-element transmission approach is proposed. In this technique, multiple adjacent transmit elements emit ultrasound pulses simultaneously with zero transmit delays. This configuration results in a focused beam, significantly increasing the acoustic pressure within the field of view (FOV), as depicted in Fig. 1(a2). The increased acoustic energy at the center of the beam enhances image quality, but the narrow beam focus might limit coverage of the entire ROI.

To address this limitation, we employ a phased-array transmitting technique that dynamically steers the beam by introducing predefined time delays (TDs) to the transmit elements. By steering the beam at various angles, energy can be distributed throughout the region of interest (ROI). The angles and the number of steering beams are calculated based on the main lobe size of each transmission, which depends on the frequency, the radius of the transducer array, and other system parameters. Figure 1(a) demonstrates the effect of steering the beam with a +20° angle delay using five consecutive transmit elements, illustrating how the energy is effectively spread across the ROI, thereby improving overall image quality.

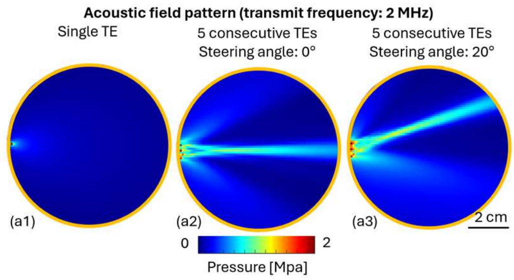

Fig. 1: Acoustic pressure fields generated by different transmission methods in RUCT systems. (a1) Single transmit element (TE) producing a broad, low-energy beam with low pressure levels across the region of interest (ROI). (a2) Five consecutive TEs transmitting simultaneously with zero steering angle, resulting in a concentrated, high-energy main lobe at the center of the ROI. (a3) Five consecutive TEs transmitting with steering angle of 20°, distributing the acoustic energy across a wider area, improving coverage and pressure distribution in the ROI, respectively.

C. Data Acquisition and Image Reconstruction

1) Data Acquisition

In both the conventional and the proposed methods, ultrasound data was acquired by sequentially transmitting ultrasound waves from the transducer elements. In the conventional method, a single element was used for each transmission. In contrast, the proposed method utilized multiple adjacent elements for each transmission, enhancing the acoustic energy delivered to the ROI.

For each transmission, a group of elements (in the proposed method) or a single element (in the conventional method) transmitted the ultrasound waves sequentially. The transmitted ultrasound waves were captured by the remaining elements, ensuring comprehensive data collection from different angles. The sampling rate for data acquisition was set at 25 MHz, providing high temporal resolution for capturing the reflected ultrasound signals.

2) Image Reconstruction

The acquired data was processed using a delay-and-sum algorithm to reconstruct the RUCT images. This algorithm aligns the received signals based on the known time delays and propagation distances, summing the aligned signals to form a coherent image [22]. To further enhance the image quality, particularly in terms of contrast and edge definition, a Hilbert transform was applied to the delay-and-sum output. The Hilbert transform facilitated the extraction of the envelope of the received signals, which corresponds to the magnitude of the acoustic pressure, thereby providing a clearer and more interpretable image.

In the proposed method, beam steering was incorporated to ensure uniform coverage of the entire ROI. The transmitted ultrasound beam was steered at multiple angles using predefined TDs applied to the transmitting elements. The data from these different steering angles were compounded to produce a single high-quality image, enhancing the SNR and improving the overall resolution.

D. Numerical Studies with Point Targets and Complex Phantom

We conducted two numerical studies using different phantoms to evaluate the performance of the proposed multi-element transmission technique compared to the conventional single-element transmission. The first study used a point target phantom, while the second study involved a more complex phantom with various geometric shapes.

1) Point Target Phantom

- **Conventional Single-Element Transmission [Fig. 2 (a1)]:** In the conventional method, a single transmit element was used for each transmission. The resulting RUCT image shows low contrast and resolution, especially in regions farther from the center, as indicated by the red and green arrows. The contrast-to-noise ratio (CNR) was measured at 28 dB for the center target (red arrow) and 21 dB for the off-center target (green arrow).

- **Multi-Element Transmission without Beam Steering [Fig. 2 (a2)]:** Using five consecutive TEs without steering significantly improved the CNR to 46 dB at the center (red arrow) but remained lower at 16 dB for the off-center target (green arrow).

- **Proposed Method with Beam Steering [Fig. 2 (a3)]:** The proposed method, which involved steering the beam at five angles (±20°, ±10°, 0°), resulted in high CNR values of

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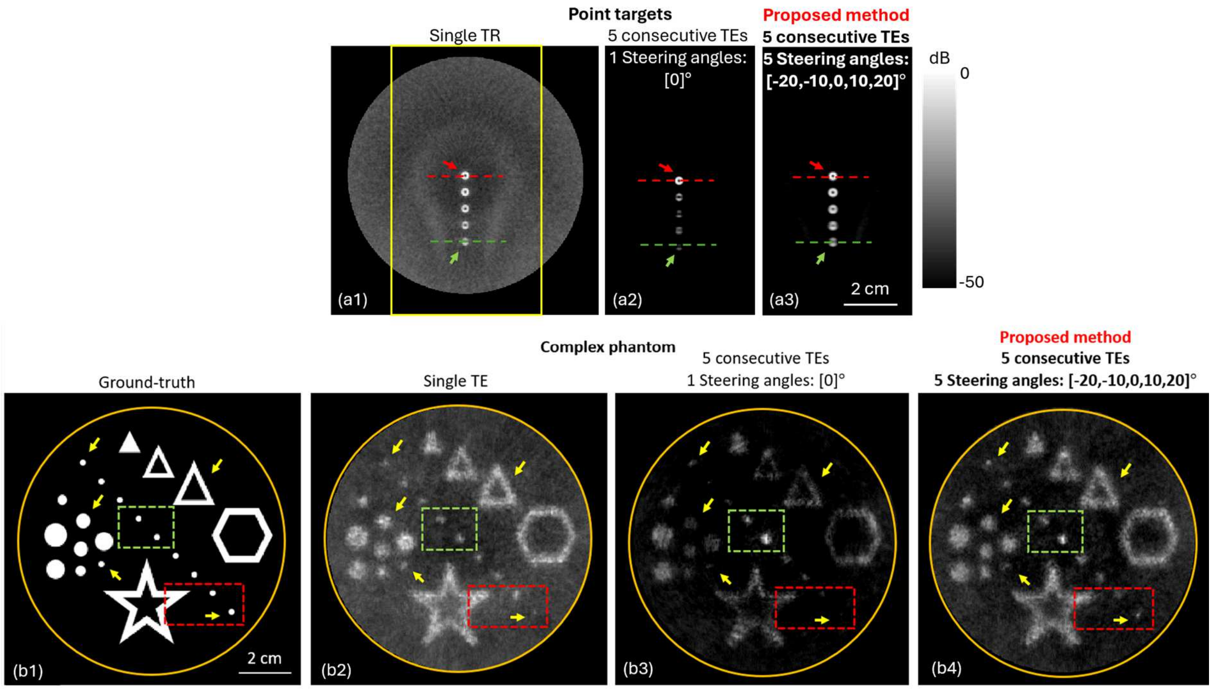

Fig. 2: Comparison of RUCT imaging techniques using different transmission methods. Top Row (b1-b3): RUCT images obtained using a point target phantom. (b1) Image produced by conventional single-element transmission. (b2) Image produced by five consecutive transmit elements (TEs) with zero steering angle, resulting in improved contrast near the center of the field of view (FOV). (b3) Image generated using the proposed method with five consecutive TEs and beam steering at angles of ±20°, ±10°, and 0°, achieving high contrast throughout the FOV. Bottom Row (c1-c4): RUCT images obtained using a complex phantom. (c1) Ground-truth image of the phantom. (c2) Image produced by conventional single-element transmission, showing significant blurring and loss of detail. (c3) Image produced by five consecutive TEs with zero steering angle, showing improved detail at the center but degradation in off-center regions. (c4) Image generated using the proposed method with beam steering, which maintains high contrast and detail across the entire FOV, closely resembling the ground-truth.

46 dB for the center and 39 dB for the off-center target, demonstrating improved image quality across the entire region of interest (ROI).

Using five TEs with zero time delays (TDs) instead of single TEs improves the image contrast near the center [as indicated by the arrows in Figure b and the green boxes in Fig. 2(a3)]. Compounding RUCT images with five steering angles results in a high-contrast image throughout the FOV, as seen by comparing the arrows and red boxes in Figure c. The proposed method enhances the CNR and SNR values by approximately 18 dB and 11 dB, respectively, compared to the conventional method [quantitative analysis shown for Figs. 2 (a2, a3)]. Achieving high-contrast harmonic images is possible because of transmitting ultrasound pulses with multiple TEs in phased-array compounding.

2) Complex Phantom

- **Conventional Single-Element Transmission [Fig. 2 (b2)]:** In the complex phantom study, the conventional method yielded an RUCT image with an SNR of 34 dB (center target) and 19 dB (off-center). The image shows blurring and loss of detail, particularly in the more complex regions of the phantom (indicated by yellow arrows).

- **Multi-Element Transmission without Beam Steering [Fig. 2 (b3)]:** The use of five consecutive TEs without beam steering enhanced the SNR to 41 dB at the center but reduced it to 11 dB for off-center targets, with significant loss of detail in complex areas.

- **Proposed Method with Beam Steering [Fig. 2 (b4)]:** The proposed method demonstrated a balanced improvement in SNR, achieving 40 dB at the center and 30 dB at off-center locations, with better preservation of detail across the entire phantom.

The method bears resemblance to techniques used in ultrasound echocardiography. By combining all RUCT images obtained at multiple transmit events, the final RUCT image achieves high contrast across the entire FOV, effectively compensating for the limitations of conventional single-element transmissions.

III. Discussion and Conclusion

This study presents a phased-array compounding technique designed to enhance the image quality of RUCT systems. By using multiple TEs with beam steering, the proposed method addresses the limitations of conventional single-element transmission, particularly the issue of low SNR at off-center locations. Our numerical studies demonstrated that this approach significantly improves both SNR and CNR across the entire FOV.

The ability to concentrate and distribute acoustic energy effectively throughout the ROI is a key advantage of this method. The phased-array compounding not only enhances the central image quality but also ensures that peripheral regions receive adequate energy, resulting in uniform and detailed imaging, even in complex phantoms. This represents a substantial improvement over traditional synthetic aperture techniques, which often struggle with low SNR in off-center regions.

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While the method offers clear benefits in terms of image quality, it introduces some challenges, such as the increased complexity in system implementation and higher computational demands for image reconstruction. The requirement for more sophisticated hardware and advanced signal processing algorithms may be a consideration for practical deployment. However, these challenges are outweighed by the significant improvements in diagnostic imaging capability.

Future work will focus on optimizing the number of TEs and steering angles to strike a balance between image quality and computational efficiency. Additionally, experimental validation on physical phantoms and eventual in vivo studies will be essential to confirm the method's applicability in clinical settings.

In conclusion, the phased-array compounding technique shows great potential in enhancing RUCT systems by delivering high-quality images across the entire FOV. Its ability to improve SNR and CNR makes it a promising advancement in ultrasound imaging, with the potential to significantly enhance diagnostic capabilities in various medical applications.

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